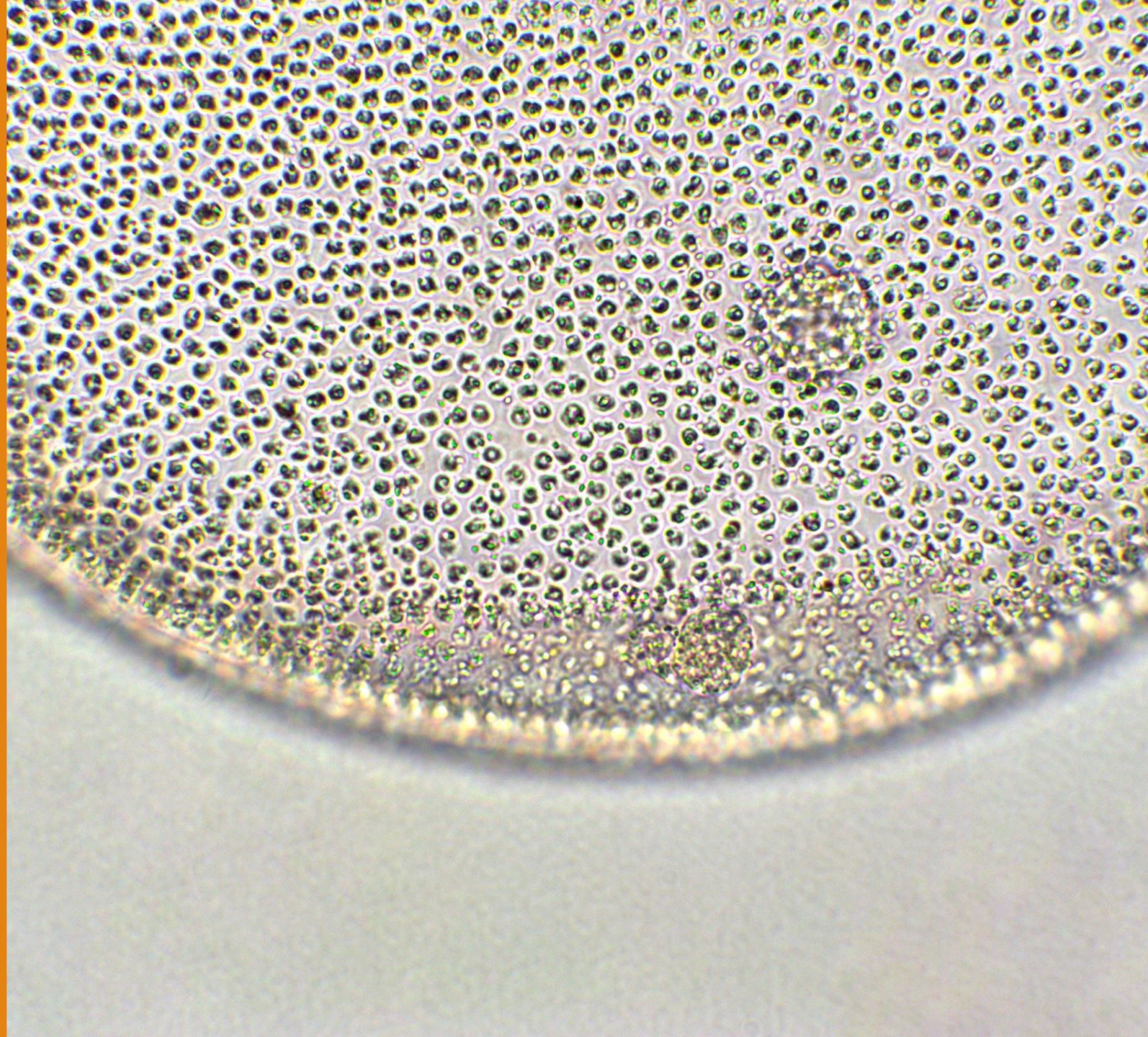


Physiology Module 1

LIVE CLASS



Branches of Physiology – matching!

1- Cell Physiology

A – How organ systems function as a team

2- Organ Physiology

B – How organs function individually

3- Systemic Physiology

C – How cells and their parts work

4- Pathophysiology

D – The study of the functions of the body

5 - Physiology

E – The study of disease's impact on the body

Location/function of subatomic particles

Subatomic Particle	Location	Charge	Function
Protons			
Neutrons			
Electrons			

Rules:

Protons: job is weight and 'name'. How many are indicated by the atomic number. They have a + charge.

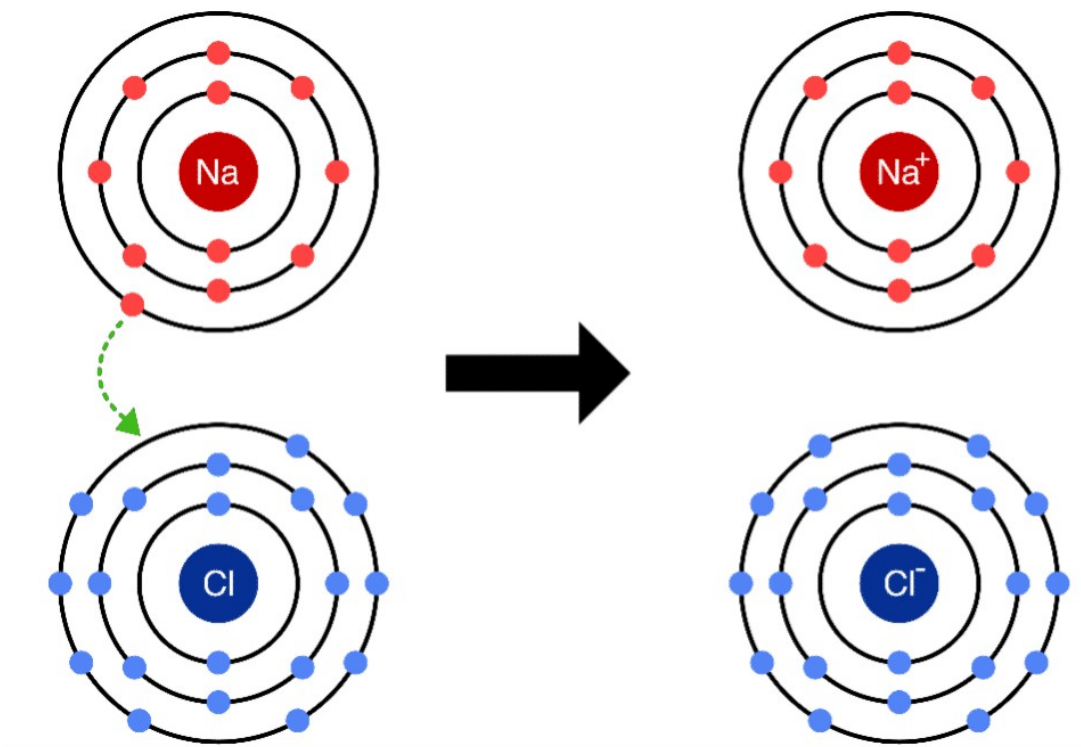
Neutrons: job is to contribute to weight. To find out how many – take the atomic mass and subtract # of protons.

$N + P = \text{weight}$ (weight – P = N)

Electrons: job is to interact with the electrons of other atoms. They have a – charge and are found in the orbits/shells.

Orbit/shell rules: First shell can hold ONLY 2 electrons, Second shell can hold up to 8 electrons, Third shell can hold up to 8 electrons. The **outer shell** is called the **Valence shell** – the electrons in the outer shell are called valence electrons.

Forming IONS



Sodium (Na) has one valence electron so it wants to give this away to another atom.

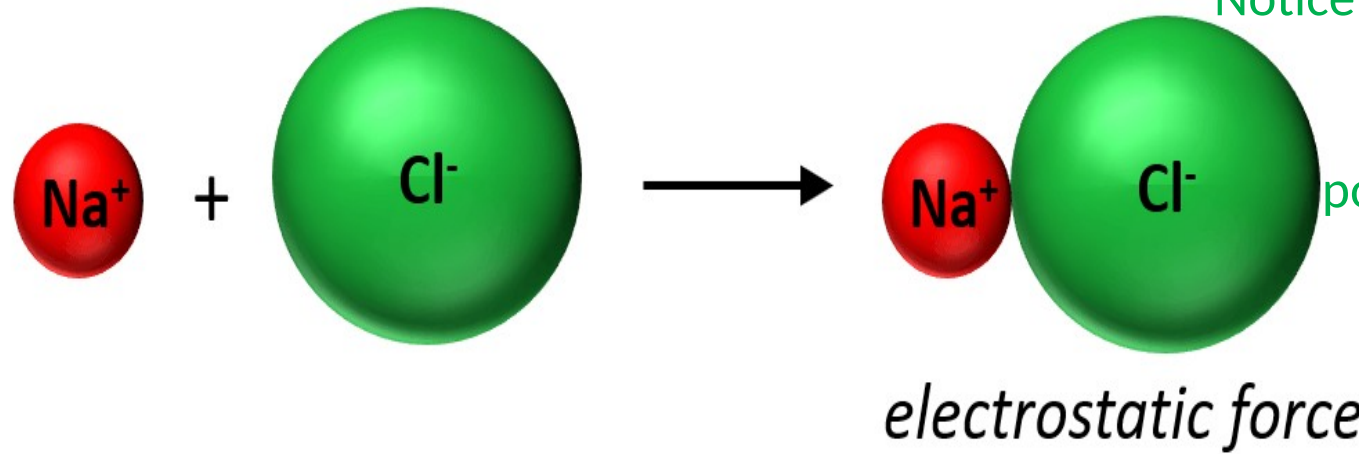
Chloride (Cl) has 7 valence electrons. It needs one more to fill the valence shell to be stable.

So these two atoms work together – sodium gives the one electron to chloride. This **creates ions!!!!**

Sodium now has more protons than electrons = more positive than negative particles so it has a positive charge = Na⁺ (a cation)

Chloride now has more electrons than protons = more negative than positive particles so it has a negative charge = Cl⁻ (an anion)

Ionic Bond

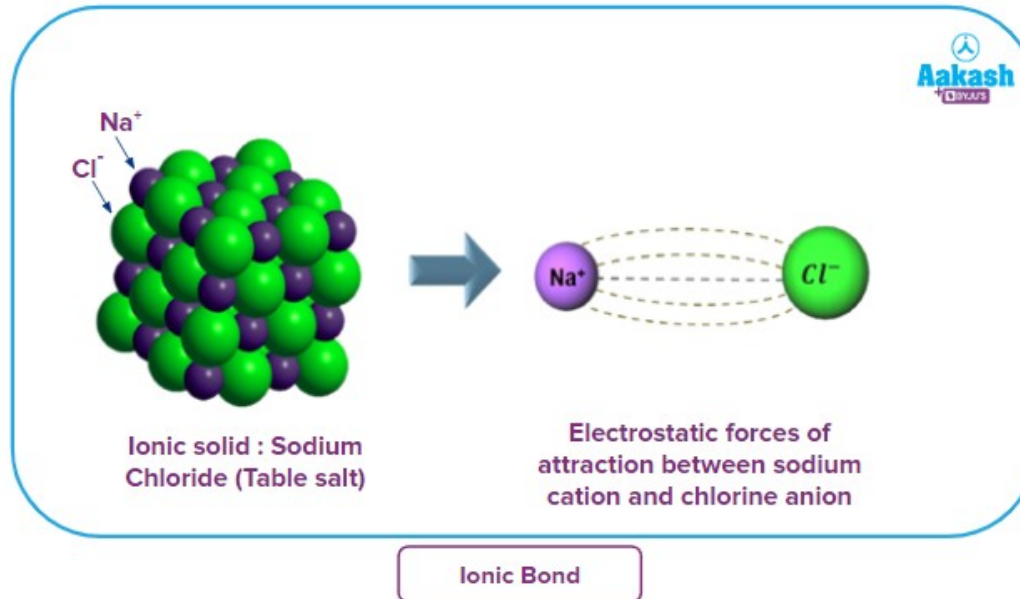


Ionic bonds form due to the attraction of opposing charges (opposites attract!)

Notice: - that these ions are no longer called atoms

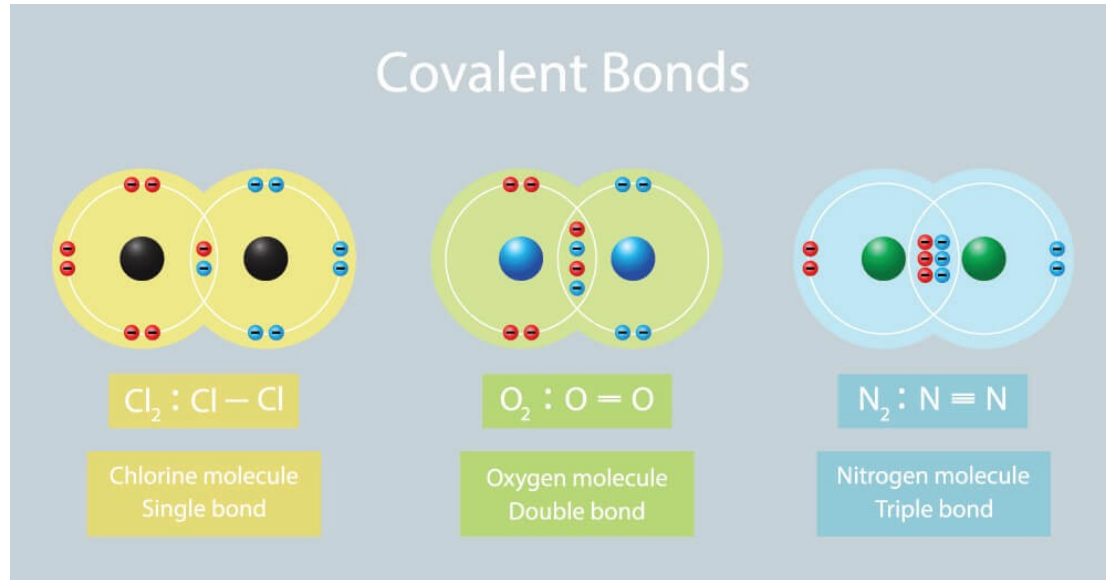
- the ions do NOT share electrons

- Ionic bond dissolve rapidly in water due to the polarity of water (coming up next)



Ionic bonds
form table salt

Covalent bonds: sharing electrons



Equal sharing = nonpolar covalent bond

Oxygen gas, nitrogen gas

Small simple no charges which means
these can diffuse through cell
membranes without any help!

Polar covalent bonding

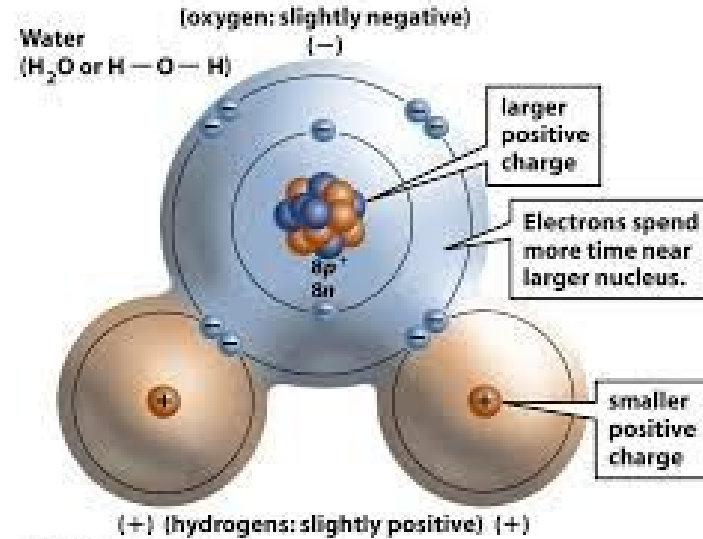
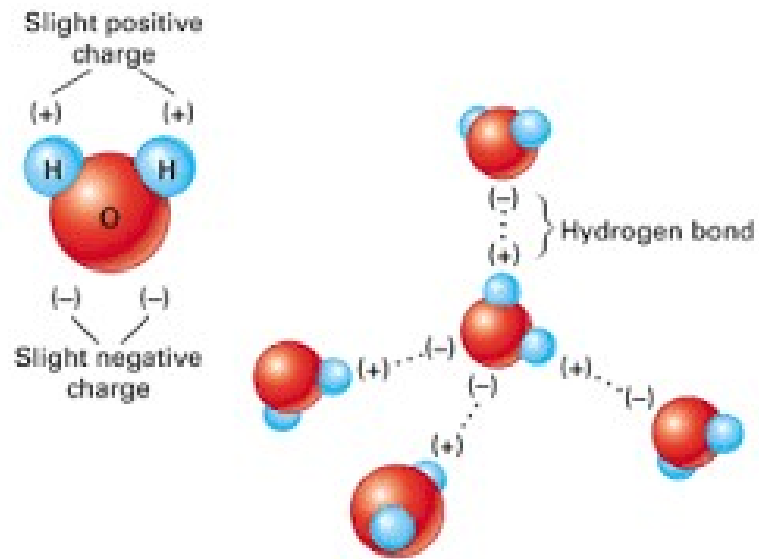


Figure 3-46 Biology (Life on Earth, 4th
© 2004 Pearson Education, Inc.)

Unequal sharing = POLAR covalent bond
Water is a great example – because it has
'poles' (slight charges) it pulls apart ionic
bonds – dissolves table salt!

Hydrogen bond – **ONLY** between polar molecules



- Hydrogen bond create an attraction between water molecules
- Hydrogen bonds form surface tension
- Hydrogen bonds hold DNA in the double helix formation

Homeostasis

The body's ability to maintain stable internal conditions even though there is continuous change in the outside environment.

Homeostatic Regulation

- **Autoregulation** is a response of organs and tissues to maintain homeostasis
 - Tissue/receptors send a message (usually a chemical) to themselves – very localized!
- **Extrinsic Regulation** requires help from the **nervous** system or **endocrine** system to maintain homeostasis.
 - * Impact is usually systemic!
 - *Nervous System and Homeostasis*: rapid, short-term, specific response.
 - *Endocrine System and Homeostasis*: chemical messengers affect tissues and organs over longer periods of time and affect growth and development
 - Pancreas is endocrine – extrinsic
 - Hypothalamus - extrinsic

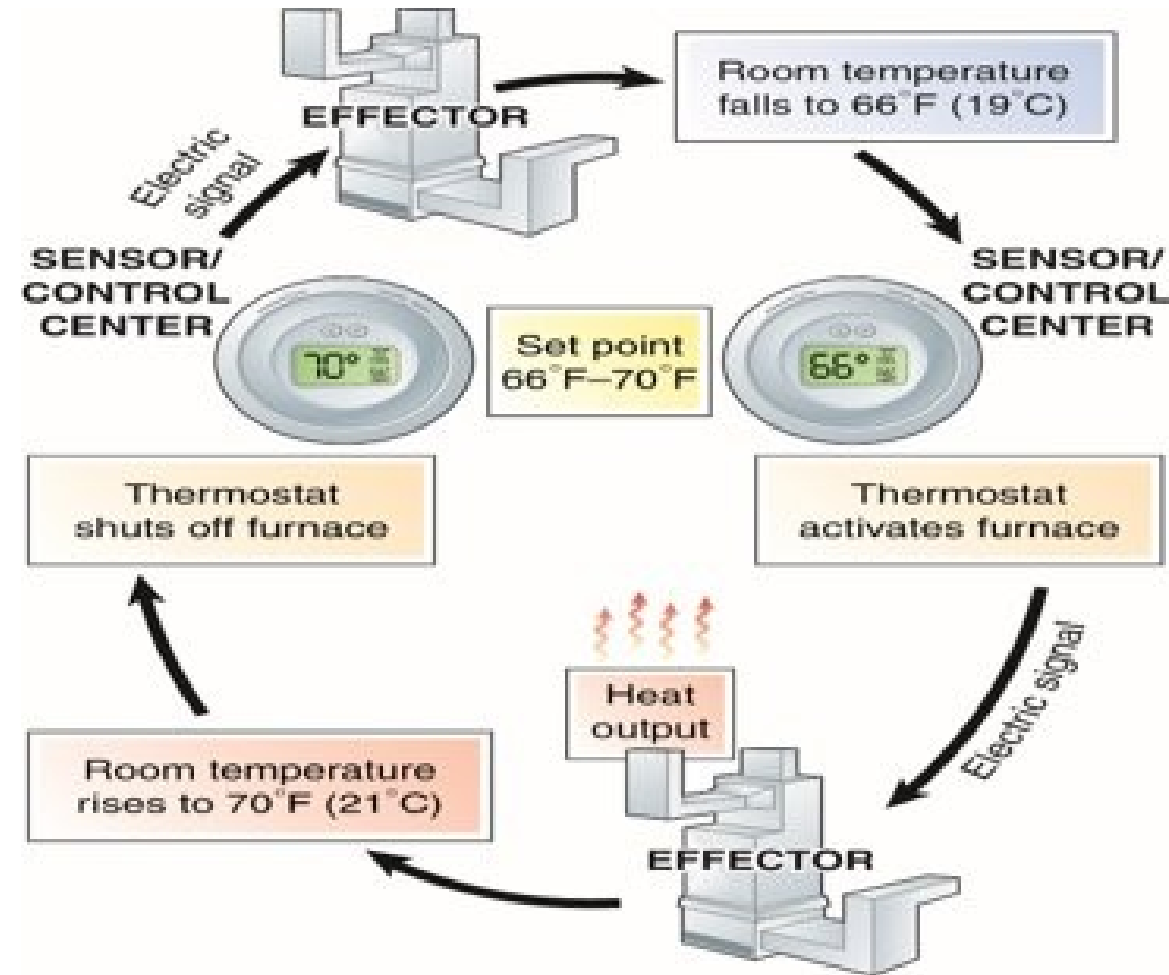
Homeostatic Regulation System

1. **receptor**: sensitive to a particular stimulus (thermometer)

2. **control center**: integration center that receives and process the information sent from the receptor; sends out commands (thermostat); establishes the set point

3. **effector**: cell or organ that responds to the control center command (air conditioner on)

Set point: **range** of desired value



Negative Feedback

The body's response *reverses or negates* the original stimulus.

This keeps the body in its normal range. Any variation outside the set point triggers the response to correct the situation until homeostasis is reached.

Has **set points – usually ranges**

Positive

A stimulus causes the initial change to continue; it ***exaggerates*** the effect rather than opposing it.

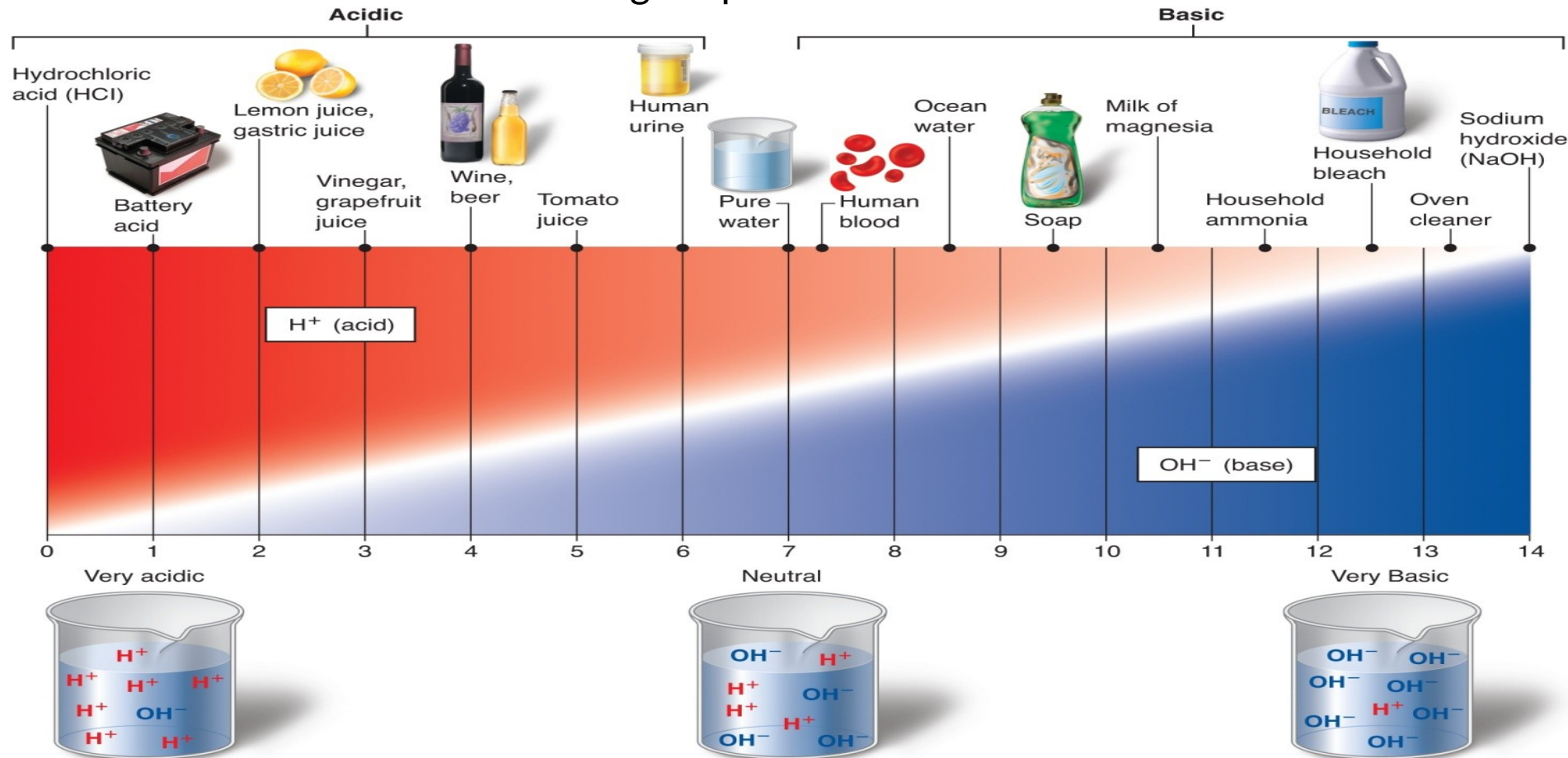
Homeostasis is achieved eventually but only when the danger, illness, or stressful situation is alleviated.

No set point – usually a **specific single goal/job**

Regulation of Blood Sugars

pH H⁺ (hydrogen ions determine pH)
MORE H⁺ = acid = lower pH value 0-7
LESS H⁺ = alkaline/base = higher pH value 7-14

Human blood: 7.35-7.45
Alkaline or acidic????



Know which
pH values are
acidic and
which are
alkaline/base.

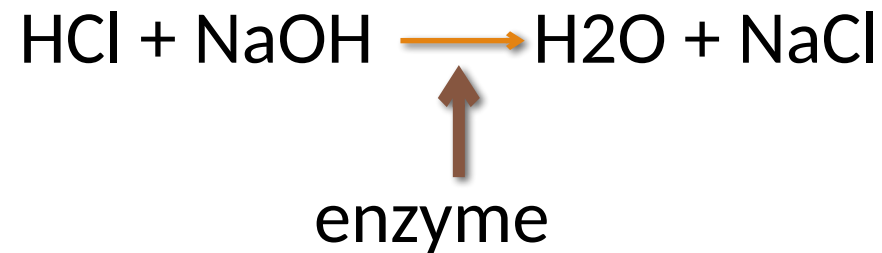
Know impact
of H⁺ amount
on pH!

Neutralizing/Buffer:

Buffers minimize pH changes!

If we combine an acid and a base, we can neutralize either and/or both:

Hydrogen Chloride (hydrochloric acid) – mixed with Sodium Hydroxide – a base:

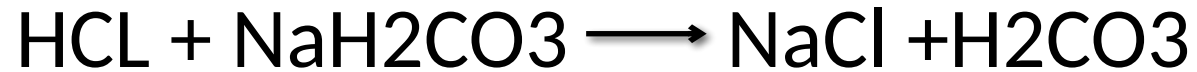


(enzymes 'help' chemical reactions happen – speed them up. More enzymes = higher product/faster.)

Buffers:

Buffers: (keeps pH changes small)

Example: CHEMICAL REACTION- acids recombine w/bases – this produces a salt and a weak acid that breaks into water and CO₂.



then when carbonic acid gets to the lungs:

H₂CO₃ breaks into H₂O + CO₂

Reminders:

Complete integrity statement for ALL shells/courses

DUE WEDNESDAY: 1st discussion post

DUE SUNDAY:

- Pre-lesson checkpoint
- Module 1 quiz (Respondus)
- Peer response in discussion
- Lab quiz (Respondus)

You can avoid feeling overwhelmed on Sunday by completing something each day – this also leads to better retention of knowledge.